

1 0. Prof's Blueprint (High Priority)

- Errors:** abs/rel error, MAE, RMSE on discrete points.
- Polynomial forms:** standard, nested (Horner), or centered; FLOP counts.
- Horner:** evaluate $p(c)$; synthetic division by $ax - b$; derivative via repeated Horner.
- Taylor:** construct poly at given center; truncation error integral & Lagrange form; bound on interval (Sandwich Thm); find n for tolerance; Maclaurin/Taylor series convergence.
- Interpolation:** setup linear system + matrix form $Xa = y$; Vandermonde det (why nonzero); over/underdetermined & free parameters; Lagrange basis $\ell_i(x)$ & poly $p(x)$; extrapolation vs interpolation; divided differences (1st/2nd order only, no full Newton); Chebyshev nodes, Runge phenomenon; Bézier curves (basic).
- MCQ + written section, 30 pts total. Bring 1 A4 cheat sheet (this one!).

2 1. Purpose of NA & Solution Types (Th.1–2)

Why numerical methods? Exact answers often *unavailable* or *impractical*: (a) no closed-form expression exists (e.g. $\int e^{-x^2} dx$, $\sin x = x$); (b) formula exists but is too slow/unstable to evaluate; (c) data only known at discrete measured points. **Definitions:** **exact solution** = satisfies equation exactly (symbolic); **closed-form/analytical** = explicit expression in known functions/constants; **approximate (numerical)** = value from an algorithm with controlled error, generally not equal to exact. **Locality:** every approximation is built to be good *near* a reference point/interval — Taylor about x_0 , interpolation on nodes $[x_0, x_n]$. Quality degrades away from that neighborhood $\Rightarrow$ **extrapolation is unreliable**.

3 2. Error Formulas

$$E_{abs} = |f(x) - \tilde{f}(x)|, \quad E_{rel} = \frac{|f(x) - \tilde{f}(x)|}{|f(x)|}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |f(x_i) - \tilde{f}(x_i)|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (f(x_i) - \tilde{f}(x_i))^2}$$

Trap: RMSE penalizes large errors more (squares); MAE treats all equally. Round to 5 dp unless told otherwise.

4 3. Polynomial Forms

Standard: $p_n(x) = a_n x^n + \dots + a_1 x + a_0$ — needs $\sim 2n - 1$ mults, n adds (naive) or $n(n+1)/2$ mults w/o precomputed powers.

Nested (Horner): $p_n(x) = (\dots((a_n x + a_{n-1})x + a_{n-2})x + \dots)x + a_0$ — n mults, n adds. **Always fewer ops.**

Centered at x_0 : $p_n(x) = \sum b_k (x - x_0)^k$. Get b_k via repeated synthetic division by $(x - x_0)$, or Taylor: $b_k = p^{(k)}(x_0)/k!$.

Horner's Method (synthetic division)

Divide $p(x)$ by $(x - c)$: bring down a_n , each next coeff = $a_k + c \cdot$ (prev result). Last value = remainder = $p(c)$.

	a_n	a_{n-1}	$\dots$	a_0
c		cb_{n-1}	$\dots$	cb_0
	b_{n-1}	b_{n-2}	$\dots$	r

- $p(x) = (x - c)q(x) + r$, $r = p(c)$; bottom row (minus last) = coeffs of $q(x)$.
- Divide by $(ax - b)$: divide by $(x - b/a)$ first (root $c = b/a$), get $q^*(x)$; true quotient $q(x) = q^*(x)/a$, same remainder r .
- Derivative link:** run the tableau on $p(x)$ to get $q(x)$ & $r = p(c)$; run *another* tableau on $q(x)$ at same $c \Rightarrow$ new remainder = $p'(c)$. (Repeat on next quotient for $p''(c)/2!$, etc.)
- Trap:** two ways to get $p'(1)$ — direct Horner on $p'(x)$, or double-tableau on $p(x)$ — must match.

5 4. Taylor & Maclaurin

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x)$$

Maclaurin: $x_0 = 0$.

Truncation Error $R_n(x)$

Integral form:

$$R_n(x) = \int_{x_0}^x \frac{(x-t)^n}{n!} f^{(n+1)}(t) dt$$

Lagrange form:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$$

for some ξ between x_0 and x . **Relationship:** Lagrange form = Mean Value Thm applied to the integral form (weighted avg of $f^{(n+1)}$ over interval).

Bounding $R_n(x)$ on interval $[a, b]$ (Sandwich Thm)

- Find $M = \max_{\xi \in [a, b]} |f^{(n+1)}(\xi)|$ (often = bound on derivative amplitude, e.g. for $\sin, \cos$ terms $M \leq 3^{n+1}$ etc.)
- Find max of $|x - x_0|^{n+1}$ on $[a, b]$ (largest distance from center).
- $|R_n(x)| \leq \frac{M \cdot d^{n+1}}{(n+1)!}$ where $d = \max |x - x_0|$.

Finding n for tolerance $|R_n| < \epsilon$

Set bound expression $< \epsilon$, solve for smallest integer n (try increasing n until factorial growth beats d^{n+1} growth). Use $n!$ growth dominates exponential — trial/error with a table is acceptable.

"Largest x " type (Quiz 3 Q5 style)

Fix n ; treat bound $\frac{M d^{n+1}}{(n+1)!} < \epsilon$ as inequality in x (since $d = x - x_0$ here); solve for x .

Series Convergence

Taylor series = $f(x) \iff \lim_{n \rightarrow \infty} R_n(x) = 0$. Writing the infinite sum from the pattern does *not* by itself prove convergence — must separately verify $R_n \rightarrow 0$ (true for $\sin, \cos, e^x$ on all $\mathbb{R}$ since M bounded, $d^{n+1}/(n+1)! \rightarrow 0$).

Maclaurin of $\sin(3x)$

$$\begin{aligned} \sin(3x) &= \sum_{k=0}^{\infty} \frac{(-1)^k (3x)^{2k+1}}{(2k+1)!} \\ &= 3x - \frac{27x^3}{6} + \frac{243x^5}{120} - \dots \end{aligned}$$

Trap: degree-8 Maclaurin poly of an odd function still stops at x^7 term (even-degree coeffs are 0) — write poly *through* degree 8, don't force an x^8 term.

6 5. Chain Rule / Derivative Practice

$[f(g(x))] = f'(g(x))g'(x)$. Common combos: $2 \sin x + e^x$; $1 + \cos^2 x = 1 + (\cos x)^2 \Rightarrow$ derivative = $-2 \cos x \sin x = -\sin(2x)$. **Worked (Quiz 2/HW):** $f = 2 \sin x + e^x$; $f' = 2 \cos x + e^x$, at $\pi/2$: $2 \cos(\pi/2) + e^{\pi/2} = e^{\pi/2} \approx 4.810$. $f = \sin^2(2x)$: $f' = 2 \sin(2x) \cos(2x) \cdot 2 = 2 \sin(4x)$ (chain $\times 2$ from $2x$). $f = \cos x - \sin x$: $f' = -\sin x - \cos x$. **Trap:** evaluate derivatives at the *center* x_0 before plugging into Taylor coeff $f^{(k)}(x_0)/k!$ — don't leave symbolic. Useful values: $\sin(\pi/2) = 1, \cos(\pi/2) = 0$; $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$; $\sin(-\pi/4) = -1/\sqrt{2}$.

7 6. Interpolation Basics

Interpolation: passes exactly through given data points (local fit). **Approximation:** fits overall trend, need not pass through points (global fit, e.g. least squares). **Extrapolation:** evaluate poly *outside* node range — much less reliable than interpolation (points 4-5 pattern in quizzes: always flag extrapolation as less trustworthy).

Degree-1 (Linear) Interpolation

Given $(x_0, y_0), (x_1, y_1)$: $p_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$.

Main Types of Interpolation (Th.8)

- Linear (piecewise):** connect consecutive points by line segments. Simple, cheap, C^0 only (kinks at nodes). Good for monotone data; crude derivative info.
- Global poly:** Lagrange/Newton/Vandermonde — one degree- n poly through $n+1$ pts; C^∞ ; **unique** for distinct nodes. High degree $\Rightarrow$ ill-conditioned, Runge.
- Cubic splines:** piecewise cubic with matched f, f', f'' at nodes; C^2 smooth, local updates, avoids Runge — best for many nodes.
- Hermite:** matches f and f' at nodes ($2n+2$ conditions $\Rightarrow$ degree $2n+1$).

Key tradeoff: more smoothness $\Rightarrow$ more conditions $\Rightarrow$ higher degree (less stable); local (spline) methods avoid Runge, global ones risk it.

Interpolation Error Bound

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

$$|f(x) - p_n(x)| \leq \frac{M}{(n+1)!} \max_x \left| \prod_{i=0}^n (x - x_i) \right|$$

where $\xi \in [x_{\min}, x_{\max}]$, $M = \max |f^{(n+1)}|$. **Uses:** bound error at a point; choose nodes to shrink product term — why Chebyshev (minimize $\max |\prod|$) beat uniform; error = 0 at every node. Analogous to Taylor remainder with $(x - x_0)^{n+1}$ replaced by $\prod (x - x_i)$.

Vandermonde System

For $n+1$ points, poly degree n : $a_0 + a_1 x_i + \dots + a_n x_i^n = y_i$ for each i . Matrix form $Xa = y$ where

$$X = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{pmatrix}$$

$$\det(X) = \prod_{0 \leq i < j \leq n} (x_j - x_i)$$

Why nonzero: interpolation nodes x_i are *distinct* $\Rightarrow$ every factor $(x_j - x_i) \neq 0 \Rightarrow \det \neq 0 \Rightarrow$ unique solution guaranteed.

Free Parameters (over/underdetermined)

points = $m+1$, poly degree = n . If $n > m$: **underdetermined** — infinitely many interpolating polys; free parameters = $n - m$ (extra coefficients not pinned down by the $m+1$ equations). If $n < m$: **overdetermined** — generally no exact solution (unless data is consistent) $\rightarrow$ use approximation (least squares) instead. **Trap:** "degree at most n " with only $m+1 < n+1$ points $\Rightarrow$ family has $n-m$ free parameters — identify them explicitly as the highest-order coefficients.

Lagrange Interpolation

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \quad p(x) = \sum_{i=0}^n y_i \ell_i(x)$$

Properties: $\ell_i(x_j) = \delta_{ij}$ (1 if $i = j$, else 0); basis *depends only on nodes x_i* , not on y_i or f — same ℓ_i reused for any function on same nodes. **Why a basis (Th.12):** $\ell_0, \dots, \ell_n$ are $n+1$ polys of degree $\leq n$ and are *linearly independent* (a combo $\sum c_i \ell_i \equiv 0$ evaluated at node x_j gives $c_j = 0$). Since $\dim P_n = n+1$, they span all polys of degree $\leq n \Rightarrow$ they form a basis; any data (x_i, y_i) yields unique coefficients $c_i = y_i$. **Steps:** (1) list nodes; (2) build each $\ell_i(x)$ as product of $(x - x_j)/(x_i - x_j)$ over $j \neq i$; (3) sum $y_i \ell_i(x)$; (4) simplify/evaluate at target point; (5) compute abs/rel error vs known true value if given; (6) extrapolate = plug x outside node range into same $p(x)$.

8 7. Divided Differences & Newton (Th.13)

0th order: $f[x_i] = f(x_i)$ (the node value itself).

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

General (order k), denom = last — first node:

$$f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Newton Interpolation Formula (Th.13)

$$p_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j)$$

Advantages vs Lagrange: adding a node adds only *one* term (no basis recompute); nested (Horner-like) eval cheaper; incremental DD table. Lagrange pros: no table, basis reusable for *any* f on same nodes. (Per prof: quizzes need only 1st/2nd order DD; know formula & why Newton wins for incremental data.)

9 8. Grids, Runge, Chebyshev, Bézier

- **Uniform grid:** equally spaced nodes, $x_i = a + ih$, $h = (b - a)/n$. **Non-uniform:** unequal spacing (e.g. Chebyshev).
- **Runge's phenomenon:** high-degree interpolation on *uniform* nodes over $[-1, 1]$ -type intervals can oscillate wildly near endpoints as n increases. Fix: Chebyshev nodes or splines.
- **Chebyshev nodes** (roots of T_{n+1}), on $[-1, 1]$:

$$x_k = \cos\left(\frac{2k+1}{2(n+1)}\pi\right), \quad k = 0, \dots, n$$

On $[a, b]$: $x_k = \frac{a+b}{2} + \frac{b-a}{2}\cos\left(\frac{2k+1}{2(n+1)}\pi\right)$. Cluster near endpoints; minimize max of $|\prod_{i=0}^n (x - x_i)|$ (interpolation error term) $\Rightarrow$ minimize Runge oscillation.

- **Bézier curves:** defined by control points; curve passes through *endpoint* control points only (not interior ones); used in computer graphics for smooth curve design; based on Bernstein polynomials as blending functions.

10 9. Quick Reference Values

$e \approx 2.71828$, $\pi \approx 3.14159$, $\sqrt{2} \approx 1.41421$, $1/\sqrt{2} \approx 0.70711$. $n!$: $5! = 120$, $6! = 720$, $7! = 5040$, $8! = 40320$, $9! = 362880$, $10! = 3628800$.

11 10. Worked Mini-Examples

Horner: $p(x) = 2x^3 - 3x^2 + 0x + 5$ at $c = 2$
Coeffs: 2, -3, 0, 5.

$$\begin{array}{r|rrrr} & 2 & -3 & 0 & 5 \\ 2 & & 4 & 2 & 4 \\ \hline & 2 & 1 & 2 & 9 \end{array}$$

So $p(2) = 9$; quotient $q(x) = 2x^2 + x + 2$; $p(x) = (x - 2)(2x^2 + x + 2) + 9$. **For** $p'(2)$: run another tableau on q 's coeffs 2, 1, 2 at $c = 2$:

$$\begin{array}{r|rrr} & 2 & 1 & 2 \\ 2 & & 4 & 10 \\ \hline & 2 & 5 & 12 \end{array}$$

So $p'(2) = 12$.

Lagrange: nodes (0, 1), (1, 3), (2, 7), **find** $p(1.5)$

$$\begin{aligned} \ell_0(x) &= \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2} \\ \ell_1(x) &= \frac{(x-0)(x-2)}{(1-0)(1-2)} = \frac{x(x-2)}{-1} \\ \ell_2(x) &= \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2} \end{aligned}$$

$p(x) = 1 \cdot \ell_0(x) + 3 \cdot \ell_1(x) + 7 \cdot \ell_2(x)$. At $x = 1.5$: $\ell_0 = \frac{(0.5)(-0.5)}{2} = -0.125$, $\ell_1 = \frac{(1.5)(-0.5)}{-1} = 0.75$, $\ell_2 = \frac{(1.5)(0.5)}{2} = 0.375$. $p(1.5) = 1(-0.125) + 3(0.75) + 7(0.375) = -0.125 + 2.25 + 2.625 = 4.75$.

Lagrange + errors: arctan (Quiz 5)

$f(x) = \arctan x$, nodes 0, 1, 3 ($y = 0, 0.785, 1.249$).

$$\begin{aligned} \ell_0(x) &= \frac{(x-1)(x-3)}{(0-1)(0-3)} = \frac{(x-1)(x-3)}{3} \\ \ell_1(x) &= \frac{x(x-3)}{(1-0)(1-3)} = \frac{x(x-3)}{-2} \\ \ell_2(x) &= \frac{x(x-1)}{(3-0)(3-1)} = \frac{x(x-1)}{6} \end{aligned}$$

$p(x) = 0\ell_0 + 0.785\ell_1 + 1.249\ell_2$. At $x = 2$: $\ell_1 = \frac{2(-1)}{-2} = 1$, $\ell_2 = \frac{2(1)}{6} = \frac{1}{3} \Rightarrow p(2) = 0.785 + \frac{1.249}{3} = 1.20133$.

Errors vs $\arctan(2) \approx 1.10715$: $E_{abs} = |1.10715 - 1.20133| = 0.09418$, $E_{rel} = \frac{0.09418}{1.10715} = 0.08507$ (8.5%).

Extrapolate $x = -0.5$ (outside $[0, 3]$): $p(-0.5) = 0.785\ell_1(-0.5) + 1.249\ell_2(-0.5) = -0.53075$ (true $\arctan(-0.5) = -0.46365$ — note large error: extrapolation unreliable).

Divided Differences: nodes (0, 1), (1, 3), (2, 7)

$f[x_0] = 1$, $f[x_1] = 3$, $f[x_2] = 7$; $f[x_0, x_1] = \frac{3-1}{1-0} = 2$, $f[x_1, x_2] = \frac{7-3}{2-1} = 4$; $f[x_0, x_1, x_2] = \frac{4-2}{2-0} = 1$. **Newton form:** $p_2(x) = 1 + 2x + 1 \cdot x(x - 1) = 1 + 2x + x^2 - x^2 + x + 1$. **Check:** same poly as Lagrange (uniqueness). At $x = 1.5$: $2.25 + 1.5 + 1 = 4.75$ ✓ matches.

Vandermonde det: nodes 0, 1, 2

$$\det(X) = \prod_{0 \leq i < j \leq 2} (x_j - x_i) = (1-0)(2-0)(2-1) = 1 \cdot 2 \cdot 1 = 2 \neq 0$$

Degree-1 Interp: $\{(2, 5), (5, -1)\}$ (Quiz 4 Q1)

$p_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0) = 5 + \frac{-1 - 5}{5 - 2}(x - 2) = 5 - 2(x - 2) = -2x + 9$. Check: $p_1(2) = 5$, $p_1(5) = -1$. ✓ (Also: slope $m = -2$ through both pts.)

Degree-4 System & det: 5 pts (Quiz 4 Q2)

Points $\{(0, 3), (1, 2), (2, 5), (4, 9), (7, 8)\}$, degree 4: $p_4(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$. **System** $Xa = y$, rows = points, cols = powers 0..4:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 4 & 16 & 64 & 256 \\ 1 & 7 & 49 & 343 & 2401 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 5 \\ 9 \\ 8 \end{pmatrix}$$

det: $\det X = \prod_{i < j} (x_j - x_i)$ over the 5 nodes (10 ordered pairs $i < j$):

$\det X = (1)(2)(1)(4)(3)(2)(7)(6)(5)(3) = 30240 \neq 0$

Solving: first eq gives $a_0 = 3$; back-substitute remaining 4 eqs (e.g. Gaussian elim) for $a_1..a_4$. Non-singular $\Rightarrow$ unique degree-4 poly.

Free Parameters: 2 points, degree ≤ 3 (Quiz 4 Q3)

Points (2, 5), (5, -1): $m + 1 = 2$, so $m = 1$; want degree $n = 3$ (4 coeffs a_0, a_1, a_2, a_3). Equations: 2. Free params = $n - m = 3 - 1 = 2 - a_2, a_3$ free. **Construct the family:** constraints $a_0 + 2a_1 + 4a_2 + 8a_3 = 5$ and $a_0 + 5a_1 + 25a_2 + 125a_3 = -1$. Subtract: $3a_1 + 21a_2 + 117a_3 = -6 \Rightarrow a_1 = -2 - 7a_2 - 39a_3$; then $a_0 = 9 + 10a_2 + 70a_3$. **Family:** $p(x) = (9 + 10a_2 + 70a_3) + (-2 - 7a_2 - 39a_3)x + a_2x^2 + a_3x^3$, any a_2, a_3 . Check: $p(2) = 5$, $p(5) = -1 \forall a_2, a_3$. □

Error calc: $\sin x \approx x$ (Quiz 1)

Exact $f(x) = \sin x$, approx $\tilde{f}(x) = x$. At $x = 0.01$: $E_{abs} = |\sin(0.01) - 0.01| \approx |0.00999983 - 0.01| = 1.67 \times 10^{-7}$; $E_{rel} = \frac{1.67 \times 10^{-7}}{\sin(0.01)} \approx 1.67 \times 10^{-5}$. **MAE** over $x = -0.01, 0.01, 0.03$: errors $1.67 \times 10^{-7}, 1.67 \times 10^{-7}, 4.50 \times 10^{-6}$; **MAE** = $\frac{1.67 + 1.67 + 45.0}{3} \times 10^{-7} \approx 1.61 \times 10^{-6}$. **RMSE**

$= \sqrt{\frac{(1.67)^2 + (1.67)^2 + (45.0)^2}{3}} \times 10^{-7} \approx 2.60 \times 10^{-6}$ (larger than MAE — squares penalize the big error).

Truncation error bound example

Generic bound formula applied literally with $n = 8$: $f(x) = \sin(3x)$, interval $[-3, 4]$, $x_0 = 0$: $f^{(9)}(x) = \pm 3^9 \cos(3x)$ or $\pm 3^9 \sin(3x) \Rightarrow M = 3^9 = 19683$. $d = \max(|-3|, |4|) = 4$.

$$|R_8(x)| \leq \frac{19683 \cdot 4^9}{9!} = \frac{19683 \cdot 262144}{362880} \approx 14219$$

(Large bound is fine — the point is applying the formula correctly, not that it's tight.)

Quiz 3 Walkthrough (truncation error)

Key: $\sin(3x)$ odd $\Rightarrow$ only odd powers; degree $= 2m + 1$ (even deg equals preceding odd).

Q1: $\sin(3x) \simeq 3x - \frac{9x^3}{2} + \frac{81x^5}{40} - \frac{243x^7}{560} + \dots = \sum_k \frac{(-1)^k 3^{2k+1} x^{2k+1}}{(2k+1)!}$ (deg 8 stops at x^7).

Q2: integral $R_{2m+1} = \frac{(-1)^{m+1} 3^{2m+2}}{(2m+1)!} \int_0^x (x-t)^{2m+1} \sin(3t) dt$;
Lagrange = $\frac{(-1)^{m+1} 3^{2m+2} \sin(3\xi)}{(2m+2)!} x^{2m+2}$.

Q3: $|R_{2m+1}| \leq \frac{12^{2m+2}}{(2m+2)!}$ on $[-3, 4]$. **Q4:**

$m = 19 \Rightarrow \frac{12^{40}}{40!} \approx 1.80 \times 10^{-5}$ (fail);

$m = 20 \Rightarrow \frac{12^{42}}{42!} \approx 1.51 \times 10^{-6} \Rightarrow m = 20$ ($2m + 2 = 42$).

Q5: even $n = 8$ invalid for odd fn $\Rightarrow n = 9$:

$\frac{3^{10} x^{10}}{10!} < 10^{-7} \Rightarrow x < 0.3012$; or $n = 7$: $\frac{3^8 x^8}{8!} \Rightarrow x < 0.1673$.

Maclaurin: $\cos(4x)$ to degree 8 (HW06)

$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \frac{u^6}{6!} + \frac{u^8}{8!} - \dots$, so with $u = 4x$:

$$\cos(4x) = 1 - 8x^2 + \frac{256}{24}x^4 - \frac{4096}{720}x^6 + \frac{65536}{40320}x^8 - \dots = 1 - 8x^2 + \frac{32}{3}x^4 - \frac{256}{45}x^6 + \frac{512}{315}x^8 - \dots$$

Pattern: $\cos(4x) = \sum_{k=0}^{\infty} \frac{(-1)^k (4x)^{2k}}{(2k)!}$; even powers only (even function), every other coefficient vanishes. $M = 4^n + 1$ (amplitude of derivative).

Taylor: $\cos x - \sin x$ at π (HW05 Q4)

$f(x) = \cos x - \sin x$, $x_0 = \pi$. Derivatives: $f' = -\sin x - \cos x$, $f'' = -\cos x + \sin x$, $f''' = \sin x + \cos x$. At π : $f(\pi) = -1$, $f'(\pi) = 1$, $f''(\pi) = 1$, $f'''(\pi) = -1$.

$$T_1 = -1 + (x - \pi)$$

$$T_2 = -1 + (x - \pi) + \frac{1}{2}(x - \pi)^2$$

$$T_3 = T_2 - \frac{1}{6}(x - \pi)^3$$

Key: each new degree adds one term; recompute all derivatives at the *center* π , divide by $k!$ (coefficients go $-1, 1, 1, -1$).

Taylor: $\sin^2(2x)$ degree 4 at 0 (HW05 Q5)

Use identity $\sin^2(2x) = \frac{1}{2}(1 - \cos 4x)$. From the $\cos(4x)$ series: $\sin^2(2x) = \frac{1}{2}\left(1 - (1 - 8x^2 + \frac{32}{3}x^4 - \dots)\right)$

$$\sin^2(2x) = 4x^2 - \frac{16}{3}x^4 + \dots \quad (\text{degree } 4)$$

Or by derivatives: $f''(0) = 8 \Rightarrow \frac{8}{2!} = 4$; $f^{(4)}(0) = -128 \Rightarrow \frac{-128}{4!} = -\frac{16}{3}$. **Trap:** $f(0) = f'(0) = f'''(0) = 0$ — don't invent odd/constant terms.

Find n for tolerance: $\cos(4x)$, $|R_n| < 0.001$ on $[-3, 2]$ (HW06 Q4)

$M = 4^{n+1}$, $d = \max|x| = 3$, bound = $\frac{4^{n+1} 3^{n+1}}{(n+1)!} = \frac{12^{n+1}}{(n+1)!}$. Increase n until bound < 0.001 :

$$\frac{12^{36}}{36!} \approx 0.00191 > 0.001, \quad \frac{12^{37}}{37!} \approx 0.00062 < 0.001$$

$\Rightarrow$ smallest n with $|R_n| < 0.001$ is $n = 36$.

Diff./Integration Quick Review

Product rule: $(fg)' = f'g + fg'$. Quotient rule:

$$(f/g)' = \frac{f'g - fg'}{g^2}. \quad \text{Chain rule: } [f(g(x))]' = f'(g(x))g'(x).$$

Integration by parts: $\int u dv =$

$uv - \int v du$. **Example:** $\int x \sin x dx$: let $u = x$, $dv = \sin x dx \Rightarrow du = dx$, $v = -\cos x$.

$$\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C$$

Centered Form: full worked example

$p(x) = x^3 - 2x^2 + 3x - 5$, center $x_0 = 2$. Repeatedly synthetic-divide by $(x - 2)$; each remainder is the next b_k (coefficient of $(x - x_0)^k$).

$$1, -2, 3, -5 \rightarrow 1, 0, 3 \mid b_0 = 1 \quad (\text{Div.1})$$

$$1, 0, 3 \rightarrow 1, 2 \mid b_1 = 7 \quad (\text{Div.2})$$

$$1, 2 \rightarrow 1 \mid b_2 = 4, b_3 = 1 \quad (\text{Div.3})$$

So $p(x) = 1 + 7(x - 2) + 4(x - 2)^2 + (x - 2)^3$. **Check:** expand back — matches original. □ Each division: bring down leading coeff, next = coeff + $2 \times$ prev; last value = remainder.

Standard $\leftrightarrow$ Nested + FLOPs (HW03/Pract.3)

$p_5(x) = x^5 + 3x^4 + x + 1$ (missing $x^3, x^2 \Rightarrow$ insert 0's). Nested:

$$p_5(x) = 1 + x(1 + x(0 + x(0 + x(3 + x))))$$

Verify: matches at $x = 0, 1, 2$. ✓

FLOPs (deg n): nested = n mult + n add = $2n$ total ops; standard (recompute each power) = mults $\frac{n(n+1)}{2}$ ($\frac{n(n-1)}{2}$ for powers + n for coeffs) + n adds = $\frac{n(n+3)}{2}$ total ops. For $n = 5$: nested = 10, standard = 20 $\Rightarrow$ **nested halves the work**.

Horner: divide by $ax - b$ (worked)

Divide $4x^3 - 7x^2 + 5x + 8$ by $2x + 1$ (i.e. $a = 2, b = -1$, root $c = -\frac{1}{2}$).

$$\begin{array}{r|rrrr} & 4 & -7 & 5 & 8 \\ -\frac{1}{2} & & -2 & 4.5 & -4.75 \\ \hline & 4 & -9 & 9.5 & 3.25 \end{array}$$

$q^*(x) = 4x^2 - 9x + 9.5$, remainder $r = 3.25$ (remainder unchanged by scaling). True quotient $q(x) = q^*(x)/2 = 2x^2 - 4.5x + 4.75$. **Check:** $(2x + 1)(2x^2 - 4.5x + 4.75) + 3.25 = 4x^3 - 7x^2 + 5x + 8$. ✓

Bézier Curves: construction

Bernstein form, n control pts $P_0, \dots, P_n$, $t \in [0, 1]$:

$$B(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i P_i$$

Quadratic ($n = 2$): $B(t) = (1-t)^2 P_0 + 2t(1-t) P_1 + t^2 P_2$. Cubic ($n = 3$): use the full Bernstein sum above. **Key facts:** curve passes through P_0 (at $t = 0$) and P_n (at $t = 1$); interior points *pull* the curve but aren't touched. **Worked example:** quadratic, $P_0 = (0, 0)$, $P_1 = (1, 2)$, $P_2 = (3, 0)$, find $B(0.5)$:

$$B(0.5) = 0.25(0, 0) + 2(0.5)(0.5)(1, 2) + 0.25(3, 0) = (0, 0) + (0.5, 1) + (0.75, 0) = (1.25, 1)$$

Common Maclaurin Series (convergence)

- $e^x = \sum \frac{x^k}{k!}$ (all $\mathbb{R}$)
- $\sin x = \sum \frac{(-1)^k x^{2k+1}}{(2k+1)!}$ (all $\mathbb{R}$)
- $\cos x = \sum \frac{(-1)^k x^{2k}}{(2k)!}$ (all $\mathbb{R}$)
- $\frac{1}{1-x} = \sum x^k$ ($|x| < 1$)
- $\ln(1+x) = \sum \frac{(-1)^{k+1} x^k}{k}$ ($|x| < 1$)
- $\tan^{-1} x = \sum \frac{(-1)^k x^{2k+1}}{2k+1}$ ($|x| \leq 1$)

Trap: writing the infinite sum doesn't prove f equals it — must show $R_n \rightarrow 0$ (see §4).